RetentionAgent: An AI-Driven Multi-Agent Framework for Employee Retention Analysis

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1. Problem Statement and Motivation

Every year, organizations lose something they cannot easily recover: their people.

According to the Gallup Global Workplace Report, employee disengagement and voluntary turnover cost the global economy approximately **$1 trillion in lost productivity annually**. Behind this number is a pattern that repeats itself across industries — a talented employee quietly disengages, considers their options, and eventually leaves. By the time a manager notices, the decision has already been made.

The core failure is not a lack of data. Most organizations already collect substantial HR data — salary records, performance reviews, tenure histories, and engagement survey scores. The failure is a lack of **early, intelligent interpretation** of that data.

Traditional HR analytics systems are built around structured, backward-looking variables. They can tell you that someone left; they struggle to tell you *why*, and they almost never tell you *before*. Three critical gaps remain systematically unaddressed:

The sentiment gap. The real drivers of employee dissatisfaction — frustration with management, a sense of career stagnation, chronic overwork — rarely appear in spreadsheets. They surface in language: in anonymous reviews, open-ended survey comments, and the gradual shift in how employees describe their work. These signals exist, but organizations lack the tools to read them at scale.

The inflation gap. A salary that looked competitive two years ago may feel inadequate today. As inflation erodes purchasing power, employees experience a real decline in compensation even when their nominal salary has not changed. Most HR systems do not account for this dynamic, leaving organizations blind to a growing source of financial dissatisfaction.

The timing gap. Even when dissatisfaction is detected, organizations typically respond too late. Retention interventions succeed only when they happen during the window between early disengagement and the decision to leave — a window that is often weeks or months wide, but only visible to systems sophisticated enough to detect its opening.

RetentionAgent is designed to close all three gaps.

2. Proposed Solution

RetentionAgent is an AI-driven employee retention analysis framework that integrates machine learning, natural language processing, economic modeling, and multi-agent reasoning into a single, cohesive analytical system.

The central design principle is that **no single signal is sufficient**. Salary data alone does not explain why a well-compensated employee is preparing to leave. Sentiment data alone cannot quantify financial risk. RetentionAgent's strength lies in combining these signals — financial, emotional, and behavioral — into a unified risk picture that is both accurate and interpretable.

Rather than producing a single turnover probability score that HR managers must interpret on their own, RetentionAgent functions like a **team of specialized analysts**, each examining a different dimension of employee experience and contributing their findings to a shared, audited conclusion. The system then translates that conclusion into specific, actionable recommendations — not just flags, but guidance on what to do next and why.

This shifts organizations from **reactive turnover management** — responding after people leave — to **proactive retention intelligence** — intervening while there is still time to act.

3. Technical Approach and Methodology

The RetentionAgent system integrates economic modeling, sentiment analysis, and multimodal machine learning to estimate employee attrition risk.

3.1 Real Wage Competitiveness Index

To measure salary competitiveness under inflation, we introduce the **Real Wage Competitiveness Index (RWCI)**:

$$RWCI = \frac{S_{internal}}{\mu_{BLS} \times (1 + \Delta CPI)}$$

Where:

- $S_{internal}$ represents the employee's internal salary
- μ_{BLS} represents the industry median wage from the U.S. Bureau of Labor Statistics
- Δ_{CPI} represents inflation derived from the Consumer Price Index

This metric measures whether an employee's compensation remains competitive relative to the external labor market after inflation adjustment.

3.2 Sentiment Shift Score

Employee sentiment is extracted from textual reviews using the **RoBERTa transformer model**. To capture emotional trend changes, we compute the **Sentiment Shift Score**:

$$S_{shift} = \frac{1}{n}\sum_{t=1}^{n}(E_t - \bar{E}_{baseline})$$

Where:

- E_t represents sentiment scores derived from employee reviews
- $\overline{E}_{baseline}$ represents the historical sentiment baseline

This metric captures deviations between recent employee sentiment and historical norms.

3.3 Multimodal Attrition Risk Modeling

Attrition risk is estimated by combining economic signals, sentiment indicators, and workload factors:

$$P_{attrition} = \sigma(\omega_1 \cdot RWCI + \omega_2 \cdot S_{shift} + \omega_3 \cdot Overtime + b)$$

Where:

- σ represents the sigmoid function
- w_i represents feature weights

This multimodal fusion model integrates **financial, emotional, and behavioral signals** into a unified attrition risk prediction.

4. Data Sources and Feasibility Considerations

The project integrates several publicly available datasets.

Internal HR Dataset

The **IBM HR Analytics Attrition Dataset** provides structured employee variables including:

- tenure
- salary
- job role
- overtime
- promotion history
- performance indicators

Salary Benchmark Data

Salary benchmarks are obtained from the **U.S. Bureau of Labor Statistics (BLS)**.

Macroeconomic Indicators

Inflation data is derived from **Consumer Price Index (CPI)** statistics to estimate real wage competitiveness.

Employee Sentiment Data

Employee sentiment signals are extracted from **Glassdoor job review datasets**.

These datasets are widely used in HR analytics research and ensure the technical feasibility of the project.

5. Project Architecture & Technical Stack

The RetentionAgent framework follows a modular architecture that integrates data engineering, machine learning models, and multi-agent reasoning. The overall system architecture is illustrated in Figure 1.

5.1 Technical Inventory (Core Components)

This section outlines the specific technologies and data structures that empower the system's analytical capabilities:

- **Multi-Source Data:** Integration of internal HR records, Glassdoor reviews (unstructured text), BLS salary benchmarks, and CPI macroeconomic indicators, all optimized in **Parquet format**.

- **Embedding & NLP Models:**
    - all-MiniLM-L6-v2: Used for transforming text into high-dimensional workplace factor scores.
    - RoBERTa: Specialized in fine-grained sentiment classification.
- **Predictive Engines:**
    - LightGBM: For high-performance nonlinear attrition modeling.
    - Cox Proportional Hazards Model: For survival analysis to determine the "timing" of employee turnover.
- **Multi-Agent Framework:** A collaborative system of specialized agents (**Emotion, Salary, Orchestrator, and Critic**) communicating via a **MongoDB blackboard architecture** for asynchronous reasoning.

5.2 System Workflow (Stages 1–5)

The following stages describe the sequence of operations, illustrating how the technical components above are orchestrated to produce actionable insights:

Stage 1 — Data Perception. The system ingests data from four sources simultaneously: Glassdoor employee reviews (unstructured text), external market salary data, internal HR records, and a structured employee attribute dataset. Raw text is separated from structured records and routed to appropriate processing pipelines.

Stage 2 — Feature Extraction. Two AI tools process the incoming data in parallel. The all-MiniLM-L6-v2 transformer model converts raw employee review text into structured dimensional scores across three workplace factors: management quality, workload pressure, and career development opportunity. Simultaneously, a LightGBM model computes the predicted salary gap — the difference between what an employee earns and what the market suggests they should earn. Both sets of outputs are stored in a MongoDB database for downstream agent access.

Stage 3 — Multi-Expert Agent Reasoning. Two specialized agents independently analyze the stored features. The **Emotion Agent** applies the RoBERTa model to sentiment feature scores, identifying patterns of emotional dissatisfaction related to management relationships, career trajectory, and workplace culture. The **Salary Agent** applies a Cox Proportional Hazards Model — a statistical technique from survival analysis — to assess the probability and timing of salary-driven attrition. Each agent produces an independent analysis report.

Stage 4 — Decision Synthesis and Audit. Both agent reports are passed to a **Decision Orchestrator**, which aggregates their findings into a comprehensive, employee-level risk assessment. Before any output is delivered, a **Critic Agent** performs a logical consistency check — verifying that recommendations are internally coherent, evidence-supported, and free from AI hallucination. This auditing layer is critical for organizational trust: HR managers need to know that AI recommendations are reliable before they act on them.

Stage 5 — Actionable Output. The verified assessment is delivered to HR managers and team leaders through a web-based dashboard, presenting employee attrition risk scores, the primary drivers of

dissatisfaction for each flagged individual, and specific recommended retention actions — whether a compensation review, a structured career conversation, or a workload adjustment. Finally, HR managers and team leaders can access these results through a centralized, web-based dashboard.

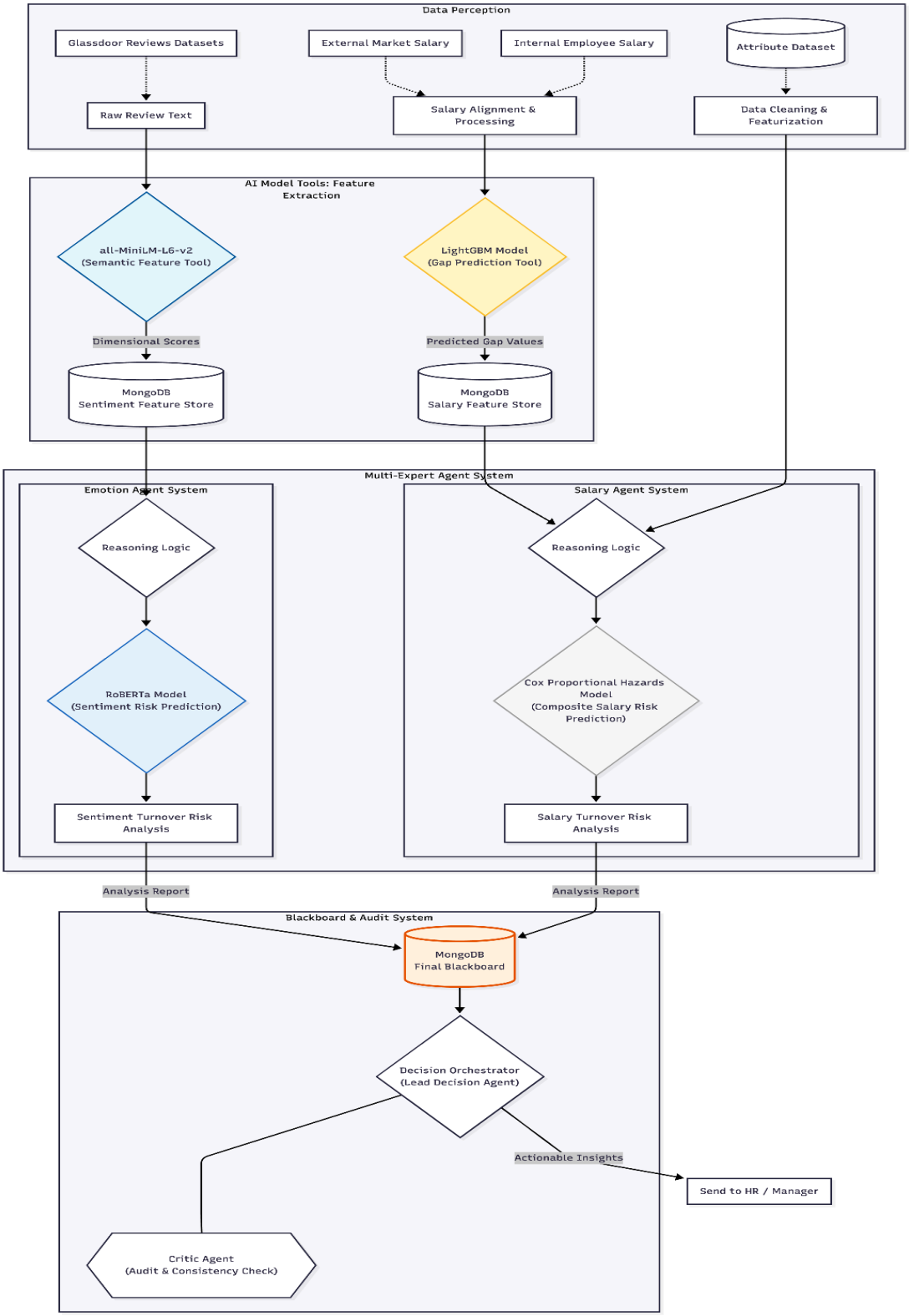

Figure 1: RetentionAgent multi-agent architecture for employee attrition risk analysis.
Click here to view the high-resolution flowchart in your browser

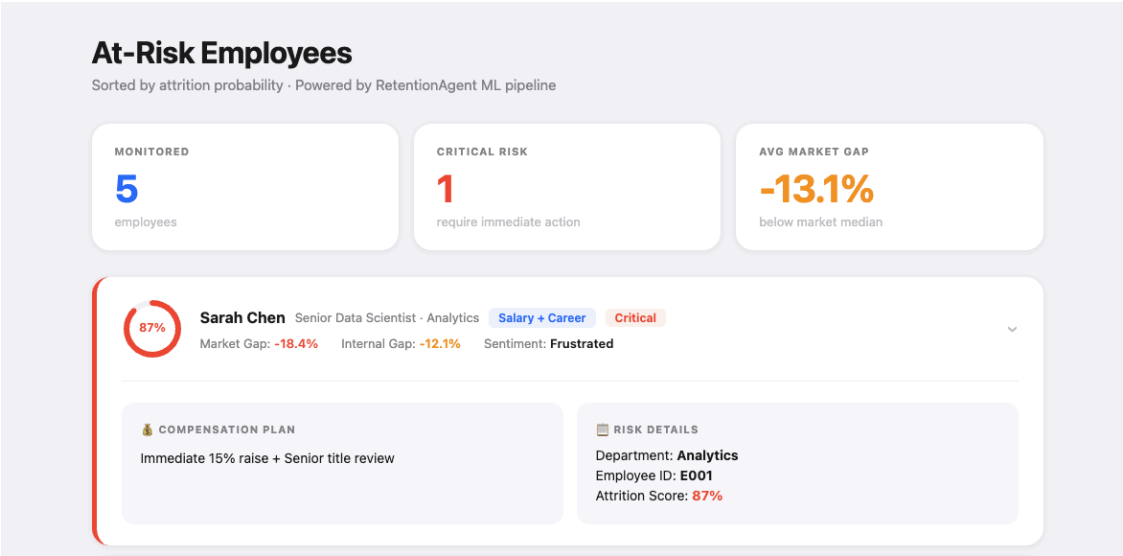

Figure 2: HR dashboard showing at-risk employees and retention recommendations.

6. Potential Impact, Use Cases, and Value Proposition

The business case for RetentionAgent is grounded in a straightforward financial reality: **it is far more expensive to replace an employee than to retain one.**

According to SHRM estimates, replacing a senior employee can cost up to 200% of their annual salary. For an engineer earning $150,000 per year, a single departure represents a replacement cost of $225,000 to $300,000 — accounting for recruitment, onboarding, lost productivity, and institutional knowledge transfer. Even modest improvements in retention rates, applied across a workforce of hundreds or thousands, generate returns that substantially exceed the cost of any analytical system designed to prevent them.

RetentionAgent delivers value across three dimensions that current solutions do not address simultaneously.

Early detection. By combining sentiment trend analysis with economic indicators, the system surfaces retention risks weeks or months before they become visible through traditional HR metrics. Organizations gain a genuine intervention window — time to act while action still matters.

Interpretability. Unlike black-box prediction models that produce a risk score without explanation, RetentionAgent identifies the specific drivers behind each employee's risk profile. HR managers do not just learn that someone is at risk; they learn whether the risk stems from salary compression, management friction, career stagnation, or workload burnout — and can respond accordingly.

Actionability. Every output from RetentionAgent is paired with a recommended intervention. The system does not stop at diagnosis; it prescribes. This closes the gap between analytical insight and organizational action — the gap where most HR analytics solutions currently fall short.

The WorkHuman sponsorship context adds a further dimension of relevance. RetentionAgent's emphasis on understanding the *human* experience behind turnover data — the frustration, the drift, the unmet expectation — aligns directly with the principle that employee retention is not purely a financial optimization problem. It is a question of whether people feel valued, heard, and supported at work.

7. Business Analysis and Commercialization Potential

RetentionAgent is designed not only as a research prototype but as the foundation for a commercially viable product.

The most natural commercialization path is a **subscription-based SaaS platform** targeting enterprise HR teams. Organizations would access a continuously updated retention risk dashboard, powered by the RetentionAgent pipeline, through a monthly or annual subscription. Integration with existing HR management systems — such as Workday or SAP SuccessFactors — would allow the platform to operate on an organization's own workforce data rather than relying on external proxies.

The total addressable market is substantial. The global HR technology market was valued at over $30 billion and continues to grow, driven by increasing organizational investment in people analytics and workforce intelligence. RetentionAgent's differentiated positioning — combining economic modeling, NLP-driven sentiment analysis, and multi-agent reasoning in a single interpretable system — addresses a capability gap that no current mainstream HR platform fully closes.

From a competitive standpoint, the system's explainability is a meaningful advantage. As organizations face increasing pressure to justify AI-driven HR decisions — particularly in regulated industries — the ability to show not just what the system recommends but why it recommends it becomes a significant differentiator.

8. Implementation Plan and Timeline

The project is structured across five phases, designed to deliver a complete, demonstrable system within the hackathon timeline.

Phase 1 — Data Preparation (Weeks 1–2). Collection and preprocessing of all datasets: IBM HR records, Glassdoor review text, BLS salary benchmarks, and CPI inflation indicators. Data is cleaned, standardized, and converted to Parquet format. Synthetic data generation is applied where needed to supplement sparse categories.

Phase 2 — Feature Engineering (Weeks 2–3). Sentiment embeddings are generated using the MiniLM transformer model. RWCI indicators are computed for each employee record. The full modeling dataset is assembled, validated, and made ready for model training.

Phase 3 — Model Development (Weeks 3–4). LightGBM models are trained for salary gap prediction. The Cox Proportional Hazards Model is implemented for survival-based attrition risk estimation.

RoBERTa is applied for sentiment classification. All models are evaluated for performance and fairness across demographic subgroups.

Phase 4 — Multi-Agent Integration (Weeks 4–5). The Emotion Agent and Salary Agent are implemented and connected. The MongoDB blackboard architecture is deployed to support asynchronous inter-agent communication. The Decision Orchestrator and Critic Agent are integrated and tested end-to-end.

Phase 5 — Evaluation and Demonstration (Weeks 5–6). Retention recommendation reports are generated and evaluated for quality and actionability. System performance is validated against held-out data. The final demonstration is prepared, including the interactive HR dashboard and a narrative walkthrough of representative employee risk cases.

References

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